Unit: 6 Unit 2: Exponents Radicals
Topic: Exponential Equations

Name: ________________
Date: ___________

Homework (Habanero)

Q1. Solve for x in the equation $2^x = 32$.
(A) $x = 3$
(B) $x = 4$
(C) $x = 5$
(D) $x = 6$
(E) $x = 7$

Q2. A certain medicine's dosage is given by $D = 5cdot2^t$, where t is the time in hours. What is the dosage after 2 hours?
(A) 20
(B) 40
(C) 50
(D) 80
(E) 100

Q3. The population of a bacteria culture grows according to the equation $P = 100cdot3^t$, where t is the time in days. What is the population after 3 days?
(A) 900
(B) $2,700$
(C) $3,000$
(D) $6,300$
(E) $9,000$

Q4. Solve for y in the equation $y = log_2(64)$.
(A) $y = 5$
(B) $y = 6$
(C) $y = 7$
(D) $y = 8$
(E) $y = 9$

Q5. A patient's white blood cell count is given by $W = 500cdot2^{0.5t}$, where t is the time in days. What is the count after 4 days?
(A) $1,000$
(B) $1,500$
(C) $2,000$
(D) $2,500$
(E) $3,000$

Q6. The probability of a certain disease is modeled by the equation $P = 1 - 2^{-x}$, where x is the number of screenings. What is the probability after 3 screenings?
(A) 0.75
(B) 0.80
(C) 0.85
(D) 0.90
(E) 0.95

Q7. The growth of a tumor is given by $G = 10cdote^{0.2t}$, where t is the time in weeks. What is the growth after 5 weeks?
(A) 50
(B) 60
(C) 70
(D) 80
(E) 90

Q8. A certain medical test's accuracy is given by $A = 80cdot(1-2^{-x})$, where x is the number of trials. What is the accuracy after 4 trials?
(A) 60
(B) 70
(C) 75
(D) 80
(E) 90

Open Ended Questions (FRQ)

Q9. Solve for x in the equation $3^{2x} = 81$ and show your work.

Q10. A patient's medication dosage is given by $D = 20cdot2^{0.5x}$, where x is the number of hours. Find the dosage after 6 hours and explain your reasoning.

Chef's Tips

- Ensure students show all steps in their work for FRQs
- Use online resources to supplement instruction
- Encourage students to ask questions and seek help when needed